Lab 3b: Advanced Cybersecurity Risk Quantification

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Task 1: Annualized Loss Expectancy (ALE)

Vulnerability	SLE (Single Loss Expectancy)	ARO	ALE
API Authentication Bypass	$5,000,000	0.15	**$750,000**
Database Injection	500,000 × $250 = **$125,000,000**	0.25	**$31,250,000**
Session Hijacking	5,000 × $1,500 = **$7,500,000**	0.40	**$3,000,000**

Formula

ALE = SLE × ARO

Task 2: Risk Prioritization Matrix

Rank	Vulnerability	SLE	ARO	ALE	Priority
1	Database Injection	$125,000,000	0.25	**$31,250,000**	Critical
2	Session Hijacking	$7,500,000	0.40	**$3,000,000**	High
3	API Authentication Bypass	$5,000,000	0.15	**$750,000**	Medium

The highest financial risk is the Database Injection Vulnerability with an ALE of $31.25 million.

Task 3

1. Bar Chart Data (ALE Comparison)

Vulnerability	ALE ($)
API Authentication Bypass	750,000
Session Hijacking	3,000,000
Database Injection	31,250,000

Chart Title: Annualized Loss Expectancy (ALE) by Vulnerability

ALE ($ Millions)



Pie Chart Data (Risk Distribution)

Total ALE = $35,000,000

Vulnerability	ALE ($)	Percentage
Database Injection	31,250,000	89.29%
Session Hijacking	3,000,000	8.57%
API Authentication Bypass	750,000	2.14%

Chart Title: Risk Distribution by ALE

Interpretation:

- Database Injection contributes **89.29%** of the organization's total risk exposure.
- Session Hijacking contributes **8.57%**.
- API Authentication Bypass contributes **2.14%**.

Risk Heat Map

Impact / Probability	Low Probability	Medium Probability	High Probability
High Impact	**API Authentication Bypass**	**Database Injection**	
Medium Impact			**Session Hijacking**
Low Impact			

Heat Map Classification

Vulnerability	Probability (ARO)	Impact (SLE)	Risk Level
API Authentication Bypass	**0.15**	**$5M**	**Medium**
Database Injection	**0.25**	**$125M**	**High**
Session Hijacking	**0.40**	**$7.5M**	**High**

ALE Bar Chart

Vulnerability	ALE ($)
API Authentication Bypass	750000
Session Hijacking	3000000
Database Injection	31250000

Task 4: Cost-Benefit Analysis

1. Advanced API Security Gateway

- Initial Cost: $350,000
- Maintenance: $50,000/year
- Total First-Year Cost: **$400,000**
- Risk Reduction: 90% × $750,000 = **$675,000**
- ROI = ((675,000 − 400,000) ÷ 400,000) × 100 = **68.75%**
- Payback Period = 400,000 ÷ 675,000 = **0.59 years**

2. Web Application Firewall (WAF)

- Initial Cost: $150,000
- Maintenance: $25,000/year
- Total First-Year Cost: **$175,000**
- Risk Reduction: 75% × $31,250,000 = **$23,437,500**
- ROI = ((23,437,500 − 175,000) ÷ 175,000) × 100 = **13,292.86%**
- Payback Period = 175,000 ÷ 23,437,500 = **0.007 years**

Multi-Factor Authentication Enhancement

- Initial Cost: $200,000
- Maintenance: $30,000/year
- Total First-Year Cost: **$230,000**
- Risk Reduction: 95% × $3,000,000 = **$2,850,000**
- ROI = ((2,850,000 − 230,000) ÷ 230,000) × 100 = **1,139.13%**
- Payback Period = 230,000 ÷ 2,850,000 = **0.08 years**

Recommended Controls

1. **Web Application Firewall (WAF)** → Highest ROI and largest risk reduction.
2. **Multi-Factor Authentication Enhancement** → Strong protection against account takeover.

Executive Summary

Top Risks

1. Database Injection: **$31.25M ALE**
2. Session Hijacking: **$3.0M ALE**
3. API Authentication Bypass: **$750K ALE**

Recommended Investment

Control	Cost (Year 1)	Risk Reduction
WAF	$175,000	$23,437,500
MFA Enhancement	$230,000	$2,850,000
API Security Gateway	$400,000	$675,000

Total Investment: $805,000
Total Risk Reduction: $26,962,500

These figures will support your charts, dashboard, heat map, ROI graphs, and executive briefing requirements in the lab.

Task 5: Control Selection Analysis

Scatter Plot: Cost vs Effectiveness

Enter this data in Excel

Control	Cost ($)	Effectiveness (%)
API Security Gateway	400,000	90
Web Application Firewall (WAF)	175,000	75
MFA Enhancement	230,000	95

Interpretation

- **MFA Enhancement** offers the highest effectiveness (95%).
- **WAF** has the lowest cost ($175,000).
- **API Security Gateway** provides strong protection but at the highest cost.

ROI Comparison Bar Chart

Control	ROI (%)
API Security Gateway	68.75
WAF	13292.86
MFA Enhancement	1139.13

Interpretation

- **WAF** delivers the highest ROI by a substantial margin.
- **MFA Enhancement** ranks second.
- **API Security Gateway** provides positive ROI but lower financial return.
- **Risk Reduction Over Time (Line Graph)**
- Assume implementation over a 12-month period.

Risk Reduction Over Time (Line Graph)

Month	Original Risk ($)	Reduced Risk ($)
0	35,000,000	35,000,000
3	35,000,000	25,000,000
6	35,000,000	15,000,000
9	35,000,000	10,000,000
12	35,000,000	8,037,500

Phase 3: Executive Reporting

Executive Summary

A quantitative risk assessment was conducted on three critical vulnerabilities within the organization. Analysis identified **Database Injection** as the most significant threat, with an Annualized Loss Expectancy (ALE) of **$31.25 million**, accounting for approximately **89.29%** of the organization's total risk exposure. **Session Hijacking** presented an ALE of **$3 million**, while **API Authentication Bypass** resulted in an ALE of **$750,000**.

Key Findings

Vulnerability	ALE ($)	Risk Ranking
Database Injection	31,250,000	1
Session Hijacking	3,000,000	2
API Authentication Bypass	750,000	3

Recommended Controls

Control	Year 1 Cost ($)	Risk Reduction ($)	ROI (%)
WAF	175,000	23,437,500	13,292.86
MFA Enhancement	230,000	2,850,000	1,139.13
API Security Gateway	400,000	675,000	68.75

Recommendations

1. Implement the **Web Application Firewall (WAF)** immediately due to its exceptionally high ROI and ability to mitigate the largest organizational risk.
2. Deploy **Multi-Factor Authentication (MFA)** to reduce account compromise and session hijacking threats.
3. Introduce the **Advanced API Security Gateway** to strengthen authentication and transaction security controls.
4. Conduct quarterly vulnerability assessments and penetration testing to verify control effectiveness.

Executive Conclusion

The assessment demonstrates that an investment of $805,000 in recommended security controls can achieve an estimated risk reduction of $26.96 million annually, providing substantial financial and operational benefits. The Web Application Firewall should be prioritized as the primary mitigation strategy due to its superior cost effectiveness and impact on overall organizational risk reduction.

Task 6: Executive Dashboard (One-Page Executive Summary)

Executive Dashboard

Overall Risk Exposure

Metric	Value
Total Annual Risk Exposure (Before Controls)	**$35,000,000**
Total Control Investment	**$805,000**
Expected Risk Reduction	**$26,962,500**
Residual Risk Exposure	**$8,037,500**

Top 3 Risks with Financial Impact

Rank	Risk	ALE ($)
1	Database Injection	31,250,000
2	Session Hijacking	3,000,000
3	API Authentication Bypass	750,000

Key Finding

Database Injection represents **89.29%** of total organizational risk exposure and should be addressed as the highest priority.

Recommended Controls and Costs

Control	Year 1 Cost ($)	Effectiveness
Web Application Firewall (WAF)	175,000	75%
Multi-Factor Authentication (MFA)	230,000	95%
API Security Gateway	400,000	90%

Expected Risk Reduction

Control	Risk Reduction ($)
WAF	23,437,500
MFA Enhancement	2,850,000
API Security Gateway	675,000

Total Risk Reduction: $26,962,500

ROI Calculations

Control	ROI (%)
WAF	13,292.86%
MFA Enhancement	1,139.13%
API Security Gateway	68.75%

Required Visualization 1

Risk Exposure Before vs After Controls (Double Bar Chart)

Data

Category	Before Controls ($)	After Controls ($)
Total Risk Exposure	35,000,000	8,037,500

Required Visualization 2

Control Investment Breakdown (Stacked Bar Chart)

Data

Control	Initial Cost ($)	Maintenance Cost ($)
WAF	150,000	25,000
MFA Enhancement	200,000	30,000
API Security Gateway	350,000	50,000

Required Visualization 3

ROI Comparison Across Controls (Horizontal Bar Chart)

Data

Control	ROI (%)
WAF	13,292.86
MFA Enhancement	1,139.13
API Security Gateway	68.75

Executive Recommendation

Priority 1

Implement the **Web Application Firewall (WAF)** immediately due to its ability to mitigate the highest-risk vulnerability and provide the strongest financial return.

Priority 2

Deploy **Multi-Factor Authentication (MFA)** to significantly reduce the risk of session hijacking and unauthorized access.

Priority 3

Implement the **API Security Gateway** to strengthen authentication mechanisms and secure transaction processing.

Executive Conclusion

The assessment of Lab 3B shows that investing **$805,000** in recommended security controls can reduce annual risk exposure from **$35 million** to **$8.04 million**, producing an estimated annual benefit of **$26.96 million**. The **Web Application Firewall** delivers the highest return on investment and should be prioritized for implementation.

Task 7: Risk Treatment Timeline (Gantt Chart)

Create the following table in Excel and convert it into a **Stacked Bar Chart** to represent a Gantt chart.

Risk Treatment Plan

Activity	Start	Duration
Deploy Web Application Firewall (WAF)	Day 1	30 Days
Configure Database Security Controls	Day 1	30 Days
Deploy Multi-Factor Authentication (MFA)	Day 31	60 Days
Security Awareness Training	Day 31	60 Days
API Security Gateway Implementation	Day 91	120 Days
Quarterly Vulnerability Assessments	Day 91	275 Days
Continuous Monitoring & Reporting	Day 91	275 Days
Annual Risk Review	Day 365	1 Day

Timeline View

Activity	0-30 Days	31-90 Days	91-365 Days
WAF Implementation	[redacted]		
Database Security Hardening	[redacted]		
MFA Deployment		[redacted]	
Security Training		[redacted]	
API Security Gateway			[redacted]
Vulnerability Assessments			[redacted]
Continuous Monitoring			[redacted]
Annual Risk Review			[redacted]

Immediate Actions (First 30 Days)

Objective: Address Highest Financial Risk

1. Implement Web Application Firewall (WAF).
2. Harden database security against injection attacks.
3. Review application input validation controls.
4. Conduct emergency vulnerability remediation.

Expected Outcome: Immediate reduction of the $31.25 million Database Injection exposure.

Short-Term Controls (31-90 Days)

1. Deploy Multi-Factor Authentication (MFA).
2. Implement stronger session management controls.
3. Conduct cybersecurity awareness training.
4. Update access control policies.

Expected Outcome: Significant reduction of Session Hijacking risk from $3 million ALE.

Long-Term Strategy (91 Days – 1 Year)

1. Deploy API Security Gateway.
2. Establish continuous monitoring and logging.
3. Conduct quarterly vulnerability assessments.
4. Perform annual enterprise risk review.

Expected Outcome: Sustainable reduction of residual risk and improved security maturity.

Executive Briefing Package

1. Completed Risk Calculations

ALE Calculations

Vulnerability	ALE ($)
Database Injection	31,250,000
Session Hijacking	3,000,000
API Authentication Bypass	750,000

Risk Prioritization Matrix

Rank	Vulnerability	ALE ($)
1	Database Injection	31,250,000
2	Session Hijacking	3,000,000
3	API Authentication Bypass	750,000

Control Analysis Worksheet

Control	Cost ($)	Risk Reduction ($)	ROI (%)
WAF	175,000	23,437,500	13,292.86
MFA	230,000	2,850,000	1,139.13
API Security Gateway	400,000	675,000	68.75

Executive Dashboard Summary

Metric	Value
Total Risk Exposure	$35,000,000
Total Investment	$805,000
Total Risk Reduction	$26,962,500
Residual Risk	$8,037,500

Final Recommendation

1. Prioritize the **Web Application Firewall (WAF)** due to its exceptionally high ROI and impact on the highest-risk vulnerability.
2. Deploy **Multi-Factor Authentication (MFA)** within the first 90 days.
3. Implement the **API Security Gateway** as part of the long-term security architecture.
4. Maintain continuous monitoring and quarterly assessments to ensure ongoing effectiveness.

Final Business Case: An investment of **$805,000** is projected to reduce annual risk exposure by **$26.96 million**, lowering total exposure from **$35 million** to **$8.04 million** and providing strong financial justification for implementation.